

Introduction, Data Sources, Data Models and Data Structures, Algorithms, DBMS, Creation of Databases (spatial and non-spatial), Spatial analysis - Interpolation, Buffer, Overlay, Terrain Modeling and Network analysis.

Surveying and Mapping. Maps: Importance of maps to engineering projects, Types of maps, Scales and uses, Plotting accuracy, Map sheet numbering, Coordinate systems- Cartesian and geographical, map projections, map datum- MSL, Geoid, spheroid, WGS-84. Land Surveying: Various Levels, Levelling methods, Compass, Theodolite and Total Station and their uses, Tachometer, Trigonometric levelling, Traversing, Triangulation and Trilateration. Aerial Photogrammetry: Types of photographs, Flying height and scale, Relief (height) displacement, Stereoscopy, 3-D Model, Height determination using Parallax Bar, Digital Elevation Model (DEM), Slope.

Image Processing and Analysis Data Quantization and Processing: Sampling and quantization theory, Principle of Linear System, Convolution, Continuous and Discrete Fourier Transform. Digital Image Processing: Digital image characteristics: image histogram and scatterogram and their significance, Variance-Covariance matrix, Correlation matrix and their significance. Radiometric and Geometric Corrections: Registration and Resampling techniques. Image Enhancement: Contrast Enhancement: Linear and Non-linear methods; Spatial Enhancement: Noise and Spatial filters . Image Transformation: Principal Component Analysis (PCA), Discriminant Analysis, Colour transformations (RGB - IHS, CMYK), Indices (Ratios, NDVI, NDWI). Image Segmentation and Classification: Simple techniques.

16. Vinod Dham Centre of Excellence for Semiconductors and Microelectronics (VDSemiX)

Discipline: Semiconductors and Microelectronics

Section 1: Computational Physics

Roots of equations, Bisection method, Regula Falsi Method, Secant, Newton-Raphson method, Convergence, Interpolation:

Finite differences and difference operators, Interpolation with equally spaced data points: Newton's forward and backward formulae for interpolation, Central difference: Gauss forward, Gauss Backward, Stirling, Bessels, Everett's formula for interpolation, Numerical solution of ordinary differential equations: Taylor's series method, Picard's method of successive approximation methods, Euler's method, modified Euler's method, Runge-Kutta method, solution of second order and simultaneous differential equations, binomial, poisson, normal and exponential distribution, Application of optimization and variational methods to problem of interest in semiconductors and microelectronics.

Section 2: Semiconductor Device Physics

Fermi energy, density of states function, quantum tunneling, potential inside a semiconductor, semiconductor in Equilibrium, Carrier transport phenomena and models, PN junction, metal semiconductor and hetero junctions, Shallow and deep levels, Carrier statistics, Carrier transport, Carrier mobility, Scattering mechanisms, Hall effect measurements, High field property, Non-equilibrium conditions, Quasi Fermi levels, Recombination processes, Current density and continuity equations, Surface recombination, Surface states, Excitons in semiconductors, Diffusion capacitance, Generation-recombination currents, Junction breakdown mechanisms. Hetero junctions: Band alignments, energy band diagrams of hetero junctions, formation of two dimensional electron gas; Metal-semiconductor contacts: Schottky barrier diode, Fermi level pinning, C-V characteristics of a Schottky diode, Current transport processes and I-V characteristics, Ohmic contacts.

Section 3: Electromagnetics

Electromagnetics; Maxwell's equations: differential and integral for $\vec{E}$ and $\vec{B}$ and the interpretation, boundary conditions, wave equation, Poynting vector; Plane waves and properties: reflection and refraction, polarization, phase and group velocity, propagation through various media, skin depth; Transmission lines: equations, characteristic impedance, impedance matching, impedance

transformation, S-parameters, Smith chart; Waveguides: modes, boundary conditions, cut-off frequencies, dispersion relations.

Section 4: Electronic Devices

Energy bands in intrinsic and extrinsic silicon; Carrier transport; diffusion current, drift current, mobility and resistivity; Generation and recombination of carriers; Poisson and continuity equations; P-N junction, Zener diode, BJT, MOS MOSFET, LED, photo diode and solar cell, Integrated circuit fabrication process: epitaxy, clean room, oxidation, diffusion, ion implantation, photo lithography, assembly techniques, packaging and twin-tub CMOS process.

Section 5: Analog System Design

Small signal equivalent circuits of diodes, BJTs and MOSFETs; Simple diode circuits; clipping, clamping and rectifiers; Single-stage BJT and MOSFET amplifiers; biasing, bias stability, mid frequency small signal analysis and frequency response; BJT and MOSFET amplifiers; multi-stage, differential, feedback, power and operational; Simple op-amp circuits; Active filters; Sinusoidal oscillators; criterion for oscillation, single-transistor and op-amp configurations; Current Mirror, Voltage and Current references, Single stage amplifiers, Cascode amplifier, Differential amplifiers, Output amplifiers, CMOS Analog opamps, Single stage and two stage Comparators, Sample and Hold circuits, Nyquist rate ADCs and DACs. Function generators, wave-shaping circuits and 555 timers; Voltage reference circuits; Power supplies; ripple removal and regulation

Section 6: Digital System Design

Number systems; Combinatorial circuits; Boolean algebra, minimization offunctions using Boolean identities and Karnaugh map, logic gates, arithmetic circuits, code converters, multiplexes, decoders, encoders, PALs and PLAs; Sequential circuits; latches and flip-flops, counters, shift-registers and finite state machines; Data converters; sample and hold circuits, ADCs and DACs; Semiconductor memories; ROM, SRAM, DRAM, Semiconductor Memory Design. Data Path subsystems– Adders, Multipliers,

Shifters, Introduction to RTL, DTL TTL, Schottky TTL, IIL and ECL logic family, concept of Noise margin, fan out and propagation delay, 16-bit microprocessor (8086); architecture, programming, memory and I/O interfacing.

Section 7: VLSI System Design

Static and dynamic characteristics of CMOS inverters, delay estimation, logical efforts and transistor sizing, power dissipation, interconnect, combinational CMOS logic circuits, complex logic circuits, behavior of MOS logic elements, SR latch circuit, clocked latch and flip-flop circuits, CMOS D-latch and edge-triggered flip-flop, pass transistor and Transmission gate logic, Stick diagram and Layout, VLSI Design Flow: Full custom and Semi Custom, Verilog/VHDL, FPGA architecture, Verification, and Testing.

AC Model of MOS, MOS resistors, current sink/ source, CMOS inverters - calculation of VTC critical points, power dissipation, rise time and fall time, Fabrication process flow, layout design rules, full custom mask layout design, stick diagram.

Section 8: Semiconductor Optoelectronics

LEDs, LASERs, Semiconductor Photo detectors, Rate of absorption in a semiconductor, materials, device structure, characteristics, power output, responsivity, speed of photo detector, PIN photo diode, APDs, Noise in Photo detectors, Applications, LIGHT MODULATORS AND PHOTOVOLTAIC DEVICES-Basic Principles, Electro-absorption Modulator: Device structure and Characteristics, Solar Cells: Device Physics, I-V characteristics in dark and light, Solar Energy Conversion, Conversion Efficiency, Materials, Applications

17. Department of Biotechnology

Molecules and their Interaction Relevant to Biology: Structure and functions of biomolecules; Carbohydrates; Fatty acids; Lipids; Amino acids; Proteins; Nucleic acids – DNA, mRNA, tRNA, rRNA; Hormones; Vitamins; Enzymes; Bio energetics; Cell metabolism; Protein-protein and protein- DNA interactions

Cellular Organization: Cell theory; Cell as basic unit of life; Hierarchy of cell organization;